Yushen Wang

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Educational Background

University of Electronic Science and Technology of China *Sept. 2022 - Jun. 2026 (expected)*

B.Eng in ***Communication Engineering***

- Rank – **2/168 (top 1.2%)** GPA – **3.92/4**
- Honors: 2024 **National Scholarship (Top 1.5%)** for Undergraduates
- Language: IELTS: **6.5** CET-4: **602** CET-6: **602**

Research Experiences

1. Lightweight Large Language Model Based on LLaMA Architecture *Nov. 2025 - Jan. 2026*

Project Description: Aimed at lightweight LLM training scenarios, this project reproduces the complete experimental pipeline from model construction, pre-training, SFT, LoRA, to RLHF, practicing the full-chain pipeline for a 64M-parameter LLM.

- **Building a Lightweight LLM from Scratch**: Implemented a lightweight model based on the Decoder-only architecture, constructing core modules including RoPE positional encoding, KV Cache, GQA, RMSNorm, FFN, and SwiGLU, providing a unified model backbone for subsequent training and alignment.
- **Complete Pre-training and Fine-tuning Workflow**: Organized the dataset construction differences across training stages such as pre-training and SFT; reproduced the pre-training and post-training code pipelines, gaining hands-on experience with mixed-precision training, gradient accumulation, checkpoint saving, resume training, and distributed training; understood dialogue template design, label construction, and loss masking to improve training stability.
- **Post-training and Preference Alignment**: Implemented reinforcement-learning-based post-training methods including PPO, DPO, and GRPO using preference data and reward signals, mastering the core ideas of how models leverage preference data, rule-based rewards, and penalty constraints to enhance reasoning capability and reasoning stability.
- **Model Evaluation and Performance Analysis**: Reproduced the lightweight LLM evaluation pipeline on C-Eval and CMMLU benchmarks, compared accuracy changes across different checkpoints, and quantitatively analyzed the impact of each training stage on Chinese knowledge QA, commonsense reasoning, and instruction-following capabilities.

2. AI Programming Assistant System Based on the ReAct Paradigm *Feb. 2026 - Mar. 2026*

Project Description: Built an AI Agent framework from scratch following the ReAct paradigm, addressing engineering challenges such as limited context windows, complex tool-calling chains, and multi-turn state maintenance, with features including multi-tool orchestration, context compression, session persistence, and long-term memory management.

- **Core Architecture Design**: Implemented an Agent loop of LLM reasoning → tool calling → result feedback, supporting multi-turn tool collaboration, output truncation recovery, and asynchronous task processing; designed a checkpoint snapshot mechanism to save states at critical tool-calling nodes, improving interrupt recovery and context consistency.
- **Plugin-style Tool System**: Built a tool registration architecture based on the Tool abstract class; new tools can be integrated simply by inheriting Tool and declaring a JSON Schema; supported concurrency-safe tiering and batch scheduling, with built-in dangerous command interception and directory sandbox restrictions in the Shell tool to reduce execution risks.
- **Tool Integration and Message Decoupling**: Implemented dynamic discovery and unified registration of external tools via the MCP Client-Server architecture, and designed an asynchronous MessageBus to decouple WebSocket interactions, the Agent main loop, and result distribution, facilitating future extensions for CLI, Cron, and third-party channels.
- **Hierarchical Memory System**: Maintained in-session dialogue context as short-term working memory with real-time token monitoring; triggered mid-term summary memory based on token thresholds—when token consumption exceeded limits, compressed historical dialogue context into structured summaries and persisted them; combined with a Dream mechanism to periodically organize cross-session long-term memory, accumulating user preferences, historical tasks, and stable facts, thereby enhancing the Agent's cross-session continuity.

Publications

- [C1] **Y. Wang**, W. Mei, X. Wei, B. Ning, and Z. Chen, "Antenna Position Optimization for Movable Antenna-Empowered Near-Field Sensing," accepted by ***ICC Workshops***, 2025, arXiv:2502.03169. [PDF]
- [J1] **Y. Wang**, W. Mei, X. Wei, B. Ning, and Z. Chen, "Movable Antenna Empowered Near-Field Sensing via Antenna Position Optimization," submitted to ***IEEE Transactions on Wireless Communications***. [PDF]